METRIC

DOD-C-64018(AR)
AMENDMENT 6
05 July 1993
SUPERSEDING
AMENDMENT 5
30 April 1991

MILITARY SPECIFICATION

CARTRIDGE, 30MM, HIGH PRESSURE TEST, HP-T, M883

This Amendment forms a part of Military Specification DOD-P-64018(AR), dated 3 October 1983, and is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.2.1, Add the following:

"9390627 - Barrel, Test: 30mm (ADEN Barrel).

"9390748 - Barrel, Test: Gun, M230, 30mm (Progressive w/6.5 deg. twist)."

PAGE 3

Delete Para 3.6 in its entirety and substitute:

"3.6 Propellant charge. The propellant charge weight used in the cartridge shall be maintained to meet the ballistic pressure requirements 3.7. The minimum charge weight required to maintain the pressure requirements of 3.7 shall be used to establish the minimum cartridge weight of 6.8".

Renumber paragraph 3.8, 3.8.1 and 3.8.2 as 3.9, 3.9.1 and 3.9.2

Add 3.8:

"3.8 Banding. The rotating band blank and the rotating band seat shall be clean and dry at the time of banding. Shot or sand blasting shall not be used to clean the band seat after knurling. The rotating band blank shall fill the band seat. When sectioned along the longitudinal axis, one half of the sectioned projectile body shall be examined for rotating band fill in the undercut areas. Failure of either of the exposed sections to comply with 65% fill in the undercut areas shall be considered a defect."

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

PAGE 7

4.4.2.1 Should read:

"203 Evidence of poor workmanship 1.5% 3.9 Visual"

PAGE 8

4.4.2.2 Should read:

"203 Protective coating damaged or incomplete 5/ 1.5% 3.1.1 Visual"

PAGE 9

Paragraph 4.4.2.2:

Change: "Projectile extraction 20 ACC-0 3.4 4.5.5"

To: "Projectile extraction 20 Acc-0 Rej-1 3.4 4.5.5/4.5.6"

Add Paragraphs 4.4.2.3, 4.4.2.4, 4.4.2.5, 4.4.2.6, 4.4.2.7, 4.4.2.8, 4.4.2.9, 4.4.2.10, 4.4.2.11, 4.4.2.12, 4.4.2.13, (Quality Conformance Inspection Table included).

PAGE 6 (9d)

* 4.4.2.5, Body and Band Assembly, Next Higher Assembly:

Delete "11825941" and substitute "9349641".

PAGE 7 (9e)

* 4.4.2.6, Projectile Nose - HPT. Drawing Number:

Delete "11825945" and substitute "12944340".

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

PAGE 10 (9h)

* 4.4.2.9, Projectile, 30mm, HPT, M883, Title:

Delete "Projectile, 30mm, HPT, M883" and substitute "Projectile Assembly".

Drawing Number: Delete "12944338" and substitute "12944339".

Next Higher Assembly: Delete "11825941" and substitute "12944338".

PAGE 11 (9i)

* 4.4.2.10, Projectile, 30mm HPT:

Drawing Number: Delete "12944339" and substitute "12944338".

Next Higher Assembly: Delete "12944338" and substitute "11825941".

PAGE 12 (9j)

* 4.4.2.11, Body and Rotating Band Assembly:

Next Higher Assembly: Delete "1299339" and substitute "12944339".

Note 1/: Delete "Each undercut area so exposed shall as minimum, be 50% filled in by the rotating band material." and substitute "Each undercut area so exposed shall as minimum, be 65% filled in by the rotating band material.".

PAGE 13 (9k)

* 4.4.2.12, Projectile Nose - HPT, M883: Delete in its entirety and substitute with attached.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile Body - HPT	1 OF 2	11825944
			NEXT HIGHER ASSEMBLY: 11825943

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined	See 4.4.2		
MAJOR				
101.	Overall length	.65	3.1	Gage
102.	Location to rotating band (from mouth)	.65	3.1	Gage
103.	Dia. bourrelet max.	.40	3.1	Gage
104.	Dia. Crimp groove max.	.40	3.1	Gage
105.	Depth, counterbore	.65	3.1	Gage
106.	Rotating band width	.65	3.1	Gage
107.	Dia. outside straight knurl	.65	3.1	Gage
108.	Presence of mouth relief	.65	3.1	Gage
109.	Foreign matter in rotating band seat	.40	3.1/3.8	Visual/3.9
110.	Dia. counterbore	.40	3.1	Gage
111.	Knurling missing or incomplete	.65	3.1	Visual

NOTES:

4(9a)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Projectile Body - HPT	2 OF 2	11825944
			NEXT HIGHER ASSEMBLY: 11825943

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
112.	Weight	.40	3.1	Scale/Balance
113.	Crimp groove location	.65	3.1	Gage
MINOR				
201.	Ogive radius location	See 4.4.2 1.0	3.1	Gage
202.	Ogive radius	1.0	3.1	Gage
203.	Distance between crimp grooves	1.0	3.1	Gage
204.	Drill depth	1.0	3.1	Gage
205.	Straight knurl depth	1.5	3.1	Gage
206.	Rotating band undercut	1.5	3.1	Gage
207.	Finish, improper	2.5	3.1	Visual
208.	Workmanship	2.5	3.9 3.9.1 3.9.2	Visual

NOTES:

5(9b)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Projectile, 30mm HPT	1 OF 1	11825942
			NEXT HIGHER ASSEMBLY: 11825941

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined	See 4.4.2		
MAJOR				
101.	Dia. rotating band maximum	.40	3.1	Gage
102.	Dia. bourrelet maximum	.40	3.1	Gage
103.	Dia. base	.40	3.1	Gage
104.	Nose gap	.65	3.1	Gage
105.	Paint adhesion	1.0	3.1	Visual
MINOR				
201.	Damaged rotating band	1.5	3.1	Visual
202.	Marking incorrect, incomplete, illegible or missing	2.5	3.1	Visual
203.	Evidence of poor workmanship	2.5	3.9 3.9.1 3.9.2	Visual

NOTES:

(6)(c)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Body & Band Assembly - HPT	1 OF 1	11825943
			NEXT HIGHER ASSEMBLY: 11825941

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined	See 4.4.2		
MAJOR				
101.	Length, mouth to rear edge of rotating band	.65	3.1	Gage
102.	Dia. bourrelet, minimum	.40	3.1	Gage
103.	Dia. rotating band	.40	3.1	Gage
104.	Runout, rotating band with datum surface	.40	3.1	Gage
MINOR				
201.	Depth and width of undercut, front and rear of rotating band, max	1.0	3.1	Profile comparator chart
202.	Gap, junction rotating band and body	1.0	3.1	chart
203.	Evidence of poor workmanship	1.5	3.9 3.9.1 3.9.2	Visual

NOTES:

7(9d)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Projectile Nose - HPT	1 OF 1	11825945
			NEXT HIGHER ASSEMBLY: 11825942

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined	See 4.4.2		
MAJOR				
101.	Dia. base minimum	.40	3.1	Gage
102.	Dia. step minimum	.40	3.1	Gage
103.	Dia. ogive profile	.65	3.1	Gage
104.	Length of ogive	.65	3.1	Gage
MINOR				
201.	Profile ogive	1.5	3.1	Gage
202.	Evidence of poor workmanship	2.5	3.9 3.9.1 3.9.2	Visual

NOTES:

8(9e)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Marking on shipping and storage container M592, 30mm cartridge	1 OF 1	11825893
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR	None defined			
MINOR				
201.	Marking misleading or unidentifiable	65%	3.1.1	Visual

NOTES:

(9(E)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Container, shipping and storage 30mm packed	1 OF 1	11825893 or 11825891 or 11825892

NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL				
1.	Link improper (anyone of the four (4) tabs of link not in groove; link inverted; link damaged; link distorted)	100%	3.1.1	Visual
MAJOR				
101.	Cartridge inverted	100%	3.1.1	Visual
102.	Incorrect number of cartridges	.40%	3.1.1	Visual
103.	Cushions, fillers, liners and separators missing	.40%	3.1.1	Visual
104.	Tubes missing	.40%	3.1.1	Visual
105.	Link orientation (LH-RH)	.40%	3.1.1	Visual
106.	Contents loose	.40%	3.1.1	Visual
MINOR				
201.	Metallic seal missing	65%	3.1.1	Visual

NOTES:

10(9)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Projectile, 30mm, HPT, M883	1 OF 1	12944338
			NEXT HIGHER ASSEMBLY: 11825941

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Nose gap	See .25%	3.1	Gage
102.	Diameter of bourrelet	4.4.2 .25%	3.1	Gage
103.	Body ogive protruding above flush with nose ogive	.25%	3.1	Visual/Manual
MINOR				
201.	Loose nose	.40%	3.1	Manual*
202.	Profile ogive	.65%	3.1	Comparator chart
203.	Workmanship	.65%	3.8	Visual

NOTES: * If the nose moves with respect to the projectile body under thumb and finger pressure applied in the lateral rotational or axial direction, the projectile shall be classified defective.

11(9h)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Projectile, 30mm HPT	1 OF 1	12944339
			NEXT HIGHER ASSEMBLY: 12944338

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined	See 4.4.2		
MAJOR				
101.	Dia. rotating band maximum	.40	3.1	Gage
102.	Dia. bourrelet maximum	.40	3.1	Gage
103.	Dia. base	.40	3.1	Gage
104.	Nose gap	.65	3.1	Gage
105.	Paint adhesion	1.0	3.1	Visual
MINOR				
201.	Damaged rotating band	1.5	3.1	Visual
202.	Marking incorrect, incomplete, illegible or missing	2.5	3.1	Visual
203.	Evidence of poor workmanship	2.5	3.9 3.9.1 3.9.2	Visual

NOTES:

12(9i)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

DOD-C-64018(AR)
AMENDMENT 6

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Body and Rotating Band Assembly Alternate design	1 OF 1	9399641
			NEXT HIGHER ASSEMBLY: 1299339

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA		REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined	See 4.4.2			
MAJOR					
101.	Dia. rotating band	See 4.4.2	.25%	3.1	Gage
102.	True position of band w/datum surface		.25%	3.1	Gage
103.	Gap, Junction rotating band and body		.25%	3.1	4.5.1.1
104.	Band tightness	20	Acc-0 Rej-1	3.1	4.5.1 1/
105.	Length, rear to band to base	See 4.4.2	.25%	3.1	Gage
MINOR					
201.	Depth and width of undercut, front and rear of rotating band, max		.40%	3.1	Profile comparator chart for minor 201 and 202
202.	Width, rotating band		.40%	3.1	
203.	Workmanship		.65%	3.8	Visual

NOTES:
1/ Projectile assemblies shall be sectioned through the longitudinal axes to expose the band fill in the undercut areas. Each undercut area so exposed shall as minimum, be 50% filled in by the rotating band material.

13(9j)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

DOD-C-64018(AR)
AMENDMENT 6

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Nose Projectile Alternate Design	1 OF 1	12944340
			NEXT HIGHER ASSEMBLY: 12944339

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Profile ogive	See 4.4.2 .25%	3.1	Profile comparator chart (complete ogive)
MINOR				
201.	Base diameter	.40%	3.1	Gage
202.	Base length	.40%	3.1	Gage
203.	Surface finish of ogive	.65%	3.1	Visual Standards
204.	Knurl missing	.65%	3.2	Visual
205.	Workmanship	.65%	3.8	Visual

NOTES:

14(9k)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Body Projectile Alternate design	1 OF 2	9399642
			NEXT HIGHER ASSEMBLY: 9399641

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Outside diameter base, min	See 4.4.2 .25%	3.1	Gage
102.	Inside dia, base, max	.25%	3.1	Gage
103.	Inside dia, middle, max	.25%	3.1	Gage
104.	Length, mouth to rear base of rotating band seat 1/	.25%	3.1	Gage
105.	Diameter, rotating band seat	.25%	3.1	Gage
106.	Knurling missing or incomplete	.25%	3.1	Visual
107.	Crimp groove profile	.25%	3.1	Comparator
108.	Foreign matter in rotating band seat	.25%	3.1	Visual 2/
109.	Base radius	.25%	3.1	Comparator
110.	Runout inside cavity diameters (3) with datum surface	.25%	3.1	Gage
111.	Length from base to rear of band seat	.25%	3.1	Comparator
112.	Weight	.25%	3.1	Scale/Balance

NOTES: 1/ Verification of band seat may be made by inspecting a minimum of two units at start of each shift and at a minimum each four hours of production or tool change.
2/ The rotating band blank and the rotating band seat shall be clean and dry at the time of banding. Shot or sandblasting shall be used to clean the band seat after knurling. The rotating band blank shall fill the band seat.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Body Projectile Alternate design	2 OF 2	9399642
			NEXT HIGHER ASSEMBLY: 9399641

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
MINOR				
201.	Width, rotating band seat	.40%	3.1	Gage
202.	Finish improper	.40%	3.1	Visual
203.	Workmanship	.65%	3.8	Visual

NOTES:

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

4.5.3 Should read:

"4.5.3 Propellant charge. The contractor shall use his standard procedure for the charge weight establishment and minimum charge weight requirements of 3.6."

4.5.6 Last sentence should read:

"See section 6 of MIL-A-48078 and 6.10 herein."

Add 4.5.7

"4.5.7 Test barrel. Unless otherwise directed by the contracting officer, the firing tests shall be conducted in the "Barrel, Test: 30mm (ADEN barrel)" Dwg. 9390627. The contracting officer may alternatively direct use of "Barrel, Test: Gun, M230, 30mm (Progressive w/6.5 deg Twist) " Dwg. 9390748.

Add 4.5.8

"4.5.8 Band tightness. After sectioning along the longitudinal axis, one half of the sectioned projectile body shall be examined for rotating band fill in the undercut areas using a .400mm No-go pin. Failure of either of the exposed sections to comply with the 65% fill requirement shall be considered a defect."

Change Paragraph 6.5c

From: "More than five areas of 15 square millimeters or less.

To: "More than five areas of 15 square millimeters or less each."

Add Paragraph 6.5d

"Total area in excess of 75 square millimeters or more."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.

* 6.8: Delete Dwg. No. "11825947" and substitute "11825941".

* Add new paragraph 6.8.1 as follows:

"6.8.1 The formula for minimum cartridge weight, when using Projectile 12944338, when assembled in accordance with Drawing 11825941 is defined as follows:

Minimum weight (grams) = 276 grams + charge weight (grams)"

6.9 Should read:

"Contamination of explosive components. Particular care should be exercised to avoid contamination of primers and propellants by grease or oil. Such contamination can adversely affect performance. Particular care should be exercised to avoid contamination of primers and propellant by dust and other foreign matter. If contamination by foreign matter or dust is evident, the quantity shall not exceed the requirements set forth in Drawing 11825946."

Add 6.11

"6.11 Retest procedures. Failure of the material sample to comply with the pressure requirements shall result in a retest. Sample size shall be as indicated in the applicable table."

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 1305-AE29)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:09Z
Check the source to verify that this is the current version before use.